

Prior-Guided Image Analysis for Bayesian Longitudinal Maize Height Phenotyping with Fixed Cameras

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Abstract

Fixed cameras could make repeated field phenotyping more accessible, but occlusion and uncertain pixel-to-length calibration can corrupt individual-image measurements. We developed a Bayesian longitudinal maize (*Zea mays* L.) height method in which predictions from earlier images guide current-image association and candidate selection before a measurement is finalized. Physical references convert image extents to centimeters, and a robust particle filter updates height, growth, and uncertainty using only images available through each time point. Independent 2021 validation comprised 132 manual heights from 33 plants in 12 camera rows; manual heights were excluded from fitting and tuning. Among six causal estimators, the particle filter gave the lowest mean absolute error (10.97 cm) and improved on EWMA, running median, and Holt by

4.01, 7.80, and 3.08 cm, with row-cluster intervals above zero. Improvements over single-frame and Gaussian filtering were 1.06 and 0.66 cm and were uncertain. Posterior 95% intervals covered 94.7% of manual heights. In simulated candidate ambiguity, using the prior during image analysis reduced latent-height RMSE from 10.5 to 6.0 cm; a matched real-image audit contained no candidate conflict. The method provides an open, physically calibrated workflow for longitudinal maize phenotyping and shows how temporal predictions can inform image analysis before measurement extraction.

Plain Language Summary. Measuring the same plants by hand throughout a field season is slow. Fixed cameras can reduce repeated field visits and plant disturbance, but leaves can overlap and an image must be converted from pixels to centimeters. We developed a method that uses what earlier images suggest about a plant’s position and height while analyzing each new image. It then updates the plant’s estimated height, growth, and uncertainty. Tests against 132 manual measurements from 33 maize plants found the lowest average error among six methods. Clear advantages were seen over three simple time-series methods, while differences from single-image and Gaussian-filter methods were small and uncertain. All code, curated images, annotations, and derived validation data are public. The approach offers a transparent route to repeated field measurements from fixed cameras and reports when the data do not distinguish competing methods.

Core Ideas

- Predictive Bayesian states guide current-image analysis before height measurement.
- Fixed cameras and physical references support longitudinal maize height phenotyping.
- Held-out 2021 manual heights independently validate the causal pipeline.
- Particle filtering improves simple longitudinal baselines; gains over stronger comparators remain uncertain.

Keywords: Bayesian longitudinal model; prior-guided image analysis; maize height; particle filter; stationary camera; uncertainty calibration

Abbreviations: CPU, central processing unit; CV, coefficient of variation; CVAT, Computer Vision Annotation Tool; EWMA, exponentially weighted moving average; MAE, mean absolute error; RGB, red–green–blue; RMSE, root mean squared error; SHA-256, 256-bit Secure Hash Algorithm; XML, Extensible Markup Language; YOLO, You Only Look Once.

1 Introduction

Longitudinal plant-height trajectories in maize (*Zea mays* L.) describe growth rate, developmental timing, and response to environment more directly than a single mature measurement. Repeated manual measurement is labor intensive and can disturb field plots, whereas fixed RGB cameras can repeatedly observe the same view with modest sensing infrastructure. Affordable phenomics depends on cost effectiveness, access, scalability, and reusable tools rather than camera price alone (Hoyos-Villegas et al., 2025). Existing time-series phenotyping systems have produced useful maize growth curves from ground and aerial images (Ge et al., 2016; Liang et al., 2018; Rodene et al., 2022; Wang et al., 2020). The practical obstacle is that a camera produces pixels rather than a physical measurement. Canopy overlap, occlusion, wind, changing illumination, and the transition from leaf tips to reproductive structures make the top and base of an individual plant ambiguous. A separate reference object can establish scale, but its geometry and usable range must be known. In a conventional two-stage pipeline, each image is reduced to a fixed measurement before the longitudinal model sees it; the downstream model cannot use earlier images to resolve the current image-level choice. A longitudinal method must also state exactly when information becomes available and how uncertainty changes as new images accumulate.

Several parts of this problem have been addressed separately. Time-lapse cameras with an in-scene scale can measure crop height, including operator-assisted rice measurements (Mano, 2017; Mano & Igawa, 2017). Stereo and structured imaging can estimate canopy height accurately, but require additional hardware or controlled acquisition (Cai et al., 2018; Tabb et al., 2019). Fixed-camera maize methods have pooled information across images, fitted nondecreasing growth curves, and used early images as labels for later overlapping plants (Guo et al., 2023; Guo et al., 2021). Other field-maize pipelines reuse locations from earlier growth stages as seed points for later segmentation or use prior-stage maize positions to guide current time-series point-cloud segmentation and height extraction (Gao et al., 2022; Li et al., 2023). These studies establish that temporal position information can guide later image analysis; they do not evaluate the same coupling of causal image analysis, physical calibration, and height-distribution updating studied here.

State-space models provide a formal way to carry information through time. Kalman and particle filters are standard tools for sequential estimation (Durbin & Koopman, 2012; Gordon et al., 1993; West & Harrison, 1997). Particle filtering has been used for real-time rice-height estimation from

83 Sentinel-1 images, and a recent Bayesian pattern-matching framework estimates daily rice height from
 84 satellite vegetation indices and sparse field measurements (Shimda et al., 2026; Yang et al., 2022). A
 85 Kalman tracker has also been combined with a maize seedling detector for counting (Li et al., 2022),
 86 while Bayesian adaptive sampling has been applied to affordable time-lapse germination phenotyping
 87 (Mercier et al., 2025). Bayesian inference necessarily uses prior information, so the mere presence of
 88 a prior is not our novelty claim. The distinction is architectural: the predictive state from earlier
 89 images is fed back into analysis of the current image before its measurement is finalized. In the field
 90 implementation, the running plant-position state guides association and base-position refinement.
 91 The stronger branch lets the predicted height distribution choose among competing top candidates;
 92 it is evaluated in controlled simulation and a matched real-image audit. Thus *prior-guided* modifies
 93 image analysis, while *Bayesian longitudinal* describes the height-growth method that supplies and
 94 updates the predictive state.

95 The study integrates three elements for longitudinal field phenotyping. First, prior-guided image
 96 analysis uses the predictive longitudinal state during current-image association and candidate selection
 97 rather than only after a measurement has been chosen. Second, in-scene physical references and
 98 recorded camera geometry convert the selected image extent to centimeters, with separate definitions
 99 for 2021 camera-support poles and 2024–2025 red-band calibration poles. Third, sequential, outlier-
 100 robust Bayesian updating returns a plant-specific height-growth distribution and exact-mixture
 101 predictive intervals. Online operation is an as-of update: when a new image is appended, the
 102 algorithm analyzes the available image prefix and reports a measurement after processing that image;
 103 later observations join the pool only when they arrive. The primary independent physical-height
 104 validation uses all available 2021 manual measurements across 33 plants, 12 stationary-camera rows,
 105 and five dates. Separate held-out tests address detector agreement, physical calibration, candidate
 106 ambiguity, and temporal transfer of predictive uncertainty.

107 2 Materials and Methods

108 2.1 Experimental and technical design

109 Figure 1 shows the information flow. Let $\mathcal{I}_{\leq t}$ be the images available through acquisition time t .
 110 The image arriving at t is appended to this pool, and the predictive state from $t - 1$ returns to
 111 image analysis before the measurement is fixed. The running position state guides association; in the

stronger branch, predicted height scores top candidates. Selected top and base rows are converted to centimeters, incorporated into the Bayesian update, and returned as an as-of distribution. Images acquired after t are absent. The 2021 study independently validates this prefix-causal model: test plants were separate, all manual labels were held out from tuning, and the five reference dates ended with the end-of-season measurement.

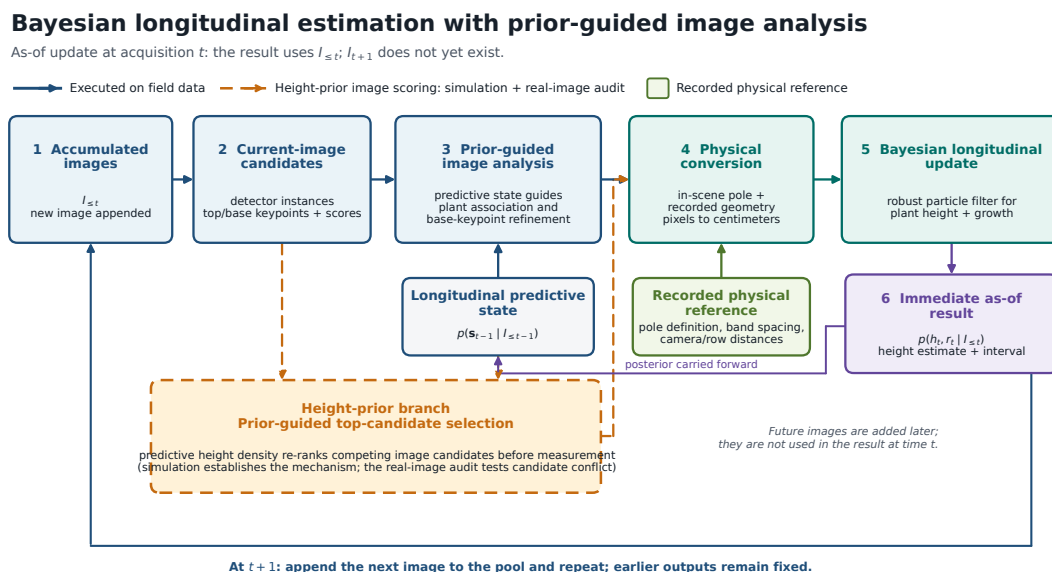


Figure 1: Workflow for the as-of image update. The feedback arrows place prior information inside current-image analysis: the running position state guides detection association and base refinement, and the dashed amber branch uses the predicted height distribution to choose among alternative top candidates. Solid blue stages were executed in the field pipeline; the height-candidate branch was evaluated in controlled simulation and the matched real-image audit. At time t , only $\mathcal{I}_{\leq t}$ is available; with settings fixed, a later image starts a new update and does not alter an earlier output.

The linked data sets have distinct roles (Table 1). Primary independent validation uses 2021 manual measurements on five dates and archived segmentation extents for all matched plant-date records across 12 stationary-camera rows; the final reference is at season end. Test plants are separate from 2024–2025 development data, and manual heights only score predictions. The 2024 development and 2025 locked-test pole images evaluate calibration, and a quality-controlled subset supplies the secondary prequential stream. Annual replanting prevents subject overlap across years, although repeated images within plants and cameras remain dependent. Separate 2021 human-annotated images support out-of-fold detector validation. Haoming Wang’s newly completed support-pole annotations describe 2021 images and audit installation geometry; they are not red-band calibration

126 truth.

Table 1: Data sources and the claim supported by each analysis. Counts are reported at the unit used for inference; boundaries prevent one evidence source from being used to support a different claim.

Data set	Units	Physical or annotation reference	Supported claim and boundary
2021 field heights	132 records; 33 plants; 12 stationary-camera rows; 5 dates	Manual height and geometrically scaled image extent	Independent physical-height accuracy; manual outcomes never fit or tune the model
2021 detector set	30 images; 150 masks	Highest and lowest pixels of each human mask	Out-of-image detector agreement; not agronomic landmark truth
2021 support poles	12 XML files; 61 retained images; 199 human traces	Nominal 5-ft support pole; ground contact to camera mount/optical-center height	Temporal reference stability; not automatic red-band accuracy
2024 pole development	7 installed cameras; 6 usable in held-out validation	Red bands spaced 1 ft edge-to-edge	Calibration development and noise estimation; no 2021 manual labels
2025 locked test	3 installed and usable calibration cameras	Same red-band definition	Cross-camera/year calibration transfer; three-camera scope
Quality-controlled sequential stream	1686 records; 48 tracks; 8 cameras; 330 images	Red-band calibration and camera geometry	Next-image interval transfer; no manual plant-height outcomes
Candidate-ambiguity simulation	480 plants in 8 scenarios	Known latent height and correct candidate	Prior-in-image mechanism under controlled ambiguity; not a field effect size

127 2.2 Physical references and geometry

128 Haoming Wang manually annotated the 2021 physical references in CVAT. The XML files contain
129 labels `pole1` through `pole4`; each label identifies a different stationary-camera support pole visible
130 in that image, and the number does not encode depth. The annotated vertical segment runs from
131 ground contact to the nominal camera mounting or optical-center height, 5 ft (152.4 cm). The
132 field records also contain four horizontal layout intervals of 35 in, totaling 140 in between adjacent
133 rows. These horizontal marks are not tick marks on the support poles. The confirmed duplicate
134 `C-039_2021-07-30.JPG.JPG` was removed while `C-039_2021-07-30.JPG` was retained. The completed
135 human audit contains 199 traces on 61 unique XML-referenced images: 172 marked usable and 27
136 marked unusable (Figure 2).

137 The 2024–2025 reference is a different object: an approximately 8–10-ft vertical calibration pole
138 with red bands separated by 1 ft (30.48 cm), measured edge-to-edge. Its visible bottom is not assumed

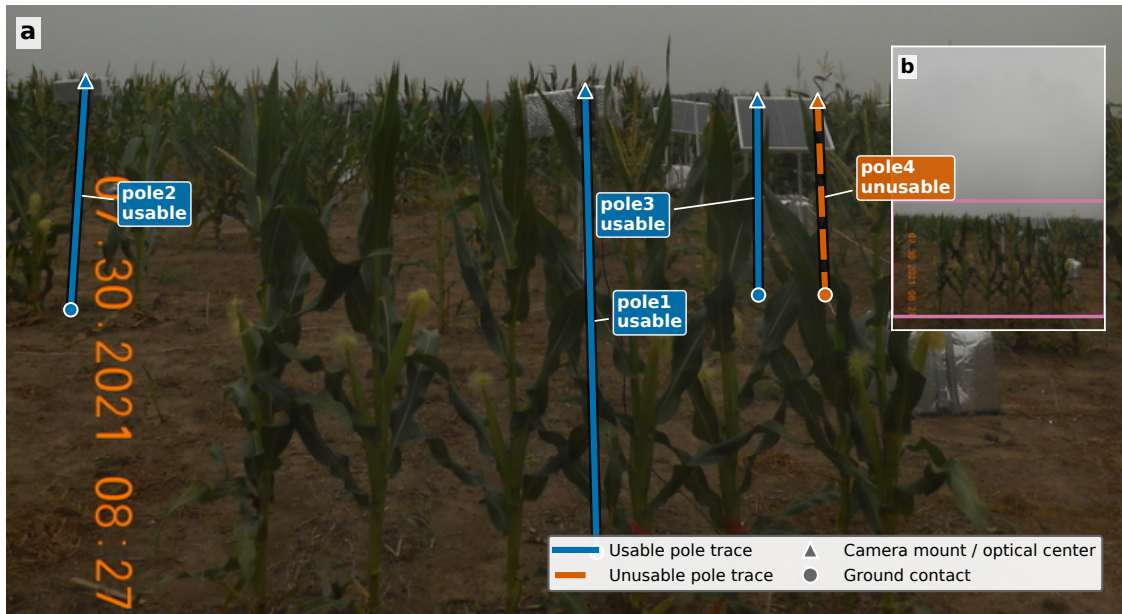


Figure 2: Representative 2021 camera-support-pole annotations in C-039_2021-07-30.JPG. Blue traces were marked usable and the orange trace unusable in the source XML. Circles denote ground contact and triangles the nominal camera mounting or optical-center height. Each traced support pole has a documented physical length of 152.4 cm. The inset shows the full retained image and crop boundary. The 35-in field marks recorded for this installation describe horizontal row layout and are not vertical pole graduations.

139 to be ground. Candidate band rows y_j are assigned relative heights z_j , and a projective map is fit
 140 within each accepted reference image,

$$z(y) = \frac{au + b}{cu + 1}, \quad u = \frac{y - \bar{y}}{s_y}. \quad (1)$$

141 Fits are rejected for insufficient bands, excessive line or height residuals, irregular spacing, nonmono-
 142 tonicity, or a near-singular denominator. Because plant height is the difference between the mapped
 143 top and base rows, the arbitrary height datum cancels.

144 Recorded 2024–2025 geometry gives a camera height of 5.0 ft, camera-to-plant-row distance
 145 $D_{\text{tgt}} = 8.5$ ft, and camera-to-reference distance $D_{\text{ref}} = 10.0$ ft. Under a pinhole camera, negligible
 146 roll, and a shared level ground plane,

$$y - y_{\text{horizon}} = f_{\text{px}} \frac{H - z}{D}. \quad (2)$$

147 Subtracting Equation 2 at the top and base shows that an extent calibrated at the pole depth is
 148 converted to the target depth by

$$h_{\text{target}} = \frac{D_{\text{tgt}}}{D_{\text{ref}}} h_{\text{pole}} = 0.85 h_{\text{pole}}. \quad (3)$$

149 Equation 3 is exact under the stated projection assumptions; uncertainty in distance, tilt, relief,
 150 lens distortion, wind, and maintenance remains empirical error. The 2021 single-frame height uses
 151 the separately documented camera geometry, 10.25 ft to the target row with a 4.5-mm focal length,
 152 6.17-mm sensor width, and 1030-pixel stored field of view, giving 0.415886 cm/pixel with zero
 153 intercept.

154 **2.3 Prior-guided image analysis within the Bayesian longitudinal method**

155 For the 2024–2025 stream, a YOLOv8-pose model produces candidate plant instances with top and
 156 root keypoints. The first complete frame initializes six left-to-right tracks. At each later frame,
 157 observed root positions are matched to running track positions by minimum total horizontal distance,
 158 subject to a maximum normalized distance of 0.16. The track position and root row are then updated
 159 with a fixed gain of 0.5. This is prior-guided image analysis because the state carried from preceding
 160 images participates in current-image association and base refinement before the physical measurement

is constructed. The sequential configuration associated 2018 detections and recovered 48 associations relative to frozen first-frame anchors; it changed the root used in the physical extent for 1963 matched records. The position recursion has the form of an exponentially weighted moving average, or a steady-state Kalman update with fixed gain. The deployed assembler does not estimate position covariance and is not a fully Bayesian neural network.

The 2021 all-plants comparison uses the archived per-image segmentation extent `height_sam`, multiplied by the recorded geometric scale. Manual field height is used only for scoring. The separate detector validation derives two keypoints from each human mask: the uppermost plant pixel and the lowermost plant pixel. This is an operational mask-extent definition, not an independently annotated flag-leaf collar. Five image-level folds were formed; a YOLOv8n-pose model was trained for 100 epochs on the other folds and evaluated on images that did not contribute to its training.

The height-prior candidate component isolates the same architectural question at the plant-top decision. Each current image in the simulation supplied one correct candidate with Gaussian error and, with probability 0.30, a competing high-confidence candidate displaced 25–60 cm upward or downward. Three estimators saw identical candidate sets: the highest-confidence single-frame candidate; the same independently selected candidate stream followed by a particle filter; and prior-guided image selection, which maximized detector confidence times the longitudinal predictive height density before filtering. The comparison therefore changes where the prior acts, not whether the downstream model is Bayesian. A complementary real-image audit used the same cached YOLO candidates, fixed geometric scale, association gates, and downstream particle filter for both temporal methods. Human-mask root positions initialized tracks only in the first available image, and manual field heights were used only for scoring. This audit comprised 63 plant-dates from 17 tracks and 6 cameras.

2.4 Bayesian filtering of longitudinal height and growth

We use *filtering* in the state-space sense: after image t is processed, the target is $p(h_t, r_t \mid Y_{1:t})$, the current height-growth distribution conditional on measurements available through t . This is distinct from spatial image filtering or denoising and from retrospective smoothing, which would use later measurements $Y_{t+1:T}$ to revise an earlier state.

For a plant observed at irregular interval Δ_t , the latent height h_t and growth rate r_t follow

$$h_t = h_{t-1} + \Delta_t r_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_h^2 \Delta_t), \quad (4)$$

$$r_t = r_{t-1} + \zeta_t, \quad \zeta_t \sim N(0, \sigma_r^2 \Delta_t), \quad (5)$$

with h_t truncated to $[0, 450]$ cm and r_t to $[-4, 15]$ cm/day. The lower growth bound permits short apparent decreases rather than forcing an irreversible monotone ratchet. The image-derived measurement Y_t has a robust two-component likelihood,

$$p(Y_t | h_t) = (1 - \pi) \phi(Y_t; h_t, \sigma_t^2) + \pi \phi(Y_t; h_t, \sigma_{\text{out}}^2), \quad (6)$$

where σ_t combines a base pose term, pole-fit residual, out-of-range distance, and band-regularity term. A bootstrap particle filter with 5000 particles propagates, weights, and resamples the state when effective sample size falls below half the particle count (Gordon et al., 1993).

Two interval targets are kept separate. A posterior interval summarizes uncertainty in latent height after image t has been incorporated. A one-step predictive interval is formed before Y_t is observed and targets the next image-derived measurement. If predicted height particles are $h_t^{(m)}$ with weights w_m , its cumulative distribution is

$$F_t(y) = \sum_m w_m \left[(1 - \pi) \Phi\left(\frac{y - h_t^{(m)}}{\sigma_t}\right) + \pi \Phi\left(\frac{y - h_t^{(m)}}{\sigma_{\text{out}}}\right) \right]. \quad (7)$$

The 0.025 and 0.975 quantiles are obtained by numerical inversion of Equation 7. This replaces the earlier Gaussian variance approximation, which did not preserve the outlier-mixture tails.

The daily 2024 development increments set $\sigma_h = 16.866$ cm/ $\sqrt{\text{day}}$, $\pi = 0.0512$, and $\sigma_{\text{out}} = 70.62$ cm from a robust median/median-absolute-deviation core-tail split over 1172 within-track intervals. Height, measurement standard deviation, process scale, and outlier scale are all expressed after the same 0.85 pole-to-target depth conversion. These values combine process and measurement variation because one observation per plant-time cannot identify those components separately. A comparison run retains the prespecified guessed values 1.5 cm/ $\sqrt{\text{day}}$, 0.08, and 60 cm, also expressed in target-row units. No 2025 observation enters the empirical noise estimates, and no 2025 plant subject appears in the 2024 development data.

For primary 2021 validation, per-image observation standard deviation ranges from 12 to 30 cm

with wind, and process-height scale is estimated from image increments. A growth-rate field $\hat{r}(h)$ learned from 2024–2025 image increments is transferred to the subject-disjoint 2021 plants. No 2021 manual height enters the image measurement, growth field, filter, noise parameters, or comparator settings. Thus the 2021 data provide independent, cross-season, label-held-out physical-height validation. The external growth field introduces no 2021 label leakage, and every update remains prefix-causal. The separate 2024-to-2025 test evaluates year-forward predictive-interval transfer.

2.5 Validation and statistical analysis

Physical calibration was evaluated by fitting alternating 1-ft automatically detected red bands and predicting the withheld bands. Cameras, rather than band records, were the sampling units for bootstrap intervals. Six 2024 cameras formed the development set and three 2025 cameras were locked until the procedure was fixed. Root-direction extrapolation was evaluated separately: the lowest bands were removed, Equation 1 was fit only to the retained upper bands, and positions 1–3 ft below the fitted range were predicted. This direction matches plant-base use; the reverse lower-to-upper experiment is not described as root extrapolation. The completed human physical-reference audit comprises the separate 2021 support-pole annotations described above; because those support poles differ from the 2024–2025 red-band objects, they document installation geometry rather than serve as truth labels for the automatic red-band candidates. Temporal repeatability was summarized for every row–pole series with at least two usable traces by the coefficient of variation of annotated pixel length. Its median and 90th percentile were accompanied by 20,000 whole-row bootstrap replicates.

Detector agreement was summarized by detection rate, keypoint error, height-extent error, and correlation in five-fold cross-fitting. The candidate-ambiguity simulation used 60 plants in each of eight misspecified growth and observation scenarios. Mean plant-level root mean squared errors and their 95% intervals were calculated by 20,000 plant bootstrap replicates. The real-image candidate audit resampled whole cameras. Secondary predictive uncertainty was evaluated at 50%, 80%, 90%, and 95% levels, together with logarithmic score and weighted interval score; settings estimated from the earlier 2024 cohort were evaluated without retuning on the temporally later, subject-disjoint 2025 cohort. Forecast rows were not treated as independent biological subjects because measurements repeat within plant tracks and cameras. The primary independent manual-reference field analysis retained every available 2021 record and compared the robust particle filter with the single-frame extent, an exponentially weighted mean, a three-image running median, a fixed Holt level-trend

240 smoother, and a Gaussian local-linear Kalman filter. Comparator settings were fixed without manual-
241 height tuning. Within-record absolute-error differences were bootstrapped 20,000 times by resampling
242 the 12 stationary-camera rows and preserving their internal dependence. Secondary robustness
243 summaries reported the complete 25th, 50th, 75th, 90th, and 95th percentiles of absolute error,
244 a whole-row bootstrap interval for the median absolute-error reduction, and the record-weighted
245 mean-error gain after omitting each camera row in turn. A mechanism-aligned secondary analysis
246 retained observations after each plant’s first image, when a preceding image was available to supply
247 longitudinal information. Central 80% and 95% latent-height posterior intervals were compared
248 with the held-out manual heights; coverage and mean width received the same whole-row bootstrap
249 treatment. Endpoint and previously residual-screened analyses are reported only as sensitivity
250 analyses in the Supplementary Materials.

251 Prefix causality was tested by rerunning every camera-time prefix and comparing all values already
252 emitted with the corresponding rows from the complete run. The audit covered 391 prefixes and
253 51,035 cumulative row comparisons.

254 **2.6 Reproducibility**

255 The public source package includes analysis scripts, fixed seeds, tabular outputs, all 61 curated 2021
256 pole-calibration images, 12 human-annotated XML pole files, the imported 199-trace audit table,
257 the released 2021 pose checkpoint with its SHA-256 checksum and recovered embedded training
258 configuration, a complete candidate cache for those images, figure sources, and manuscript source. The
259 checkpoint records the software version, seed, deterministic setting, training arguments, and summary
260 metrics; the referenced dataset file, exact training-image manifest, original training masks, and source
261 commit were not recovered. Source-frame matching recovered coordinates and match scores for 87
262 legacy plant crops; 63 join the 132-record primary validation across six camera rows, leaving 69
263 primary extents without a reconstructable raw-image chain. Headline numbers are regenerated from
264 canonical comma-separated-value or JavaScript Object Notation outputs rather than transcribed.
265 The daily 2024–2025 raw-image archive remains too large for this release, so its published analyses
266 start from checksum-identified derived measurements.

2.7 Practical deployment and transfer

The workflow requires a fixed RGB view, an in-scene physical reference, recorded camera-to-reference and camera-to-target geometry, image time and camera identity, and an initialized plant association. For each arriving image it returns the chosen plant extent, a height in centimeters, and posterior summaries for height and growth. The released pose checkpoint supports CPU inference, and the released 61-image candidate cache was generated at 640-pixel inference size in the documented software environment. Processing time, equipment purchase price, installation labor, and maintenance cost were not recorded consistently, so the study does not claim a formal cost advantage. Its accessibility rests on modest sensing requirements, open analysis code, and auditable intermediate outputs. Transfer to another camera layout or crop requires local confirmation of the physical reference and landmark definition; the 0.85 depth factor must not be reused unless the recorded geometry matches Equation 3.

3 Results

3.1 Physical calibration and extrapolation

The 172 usable 2021 support-pole traces formed 39 repeated row-pole series across all 12 stationary-camera rows. Median within-series pixel-length coefficient of variation was 1.24% (95% whole-row bootstrap interval, 0.78–1.99%), and its 90th percentile was 3.65%. 38 of 39 series were below 5%; the remaining series reached 11.51%. This audit supports temporal stability of the projected references and their annotations; it does not establish absolute scale accuracy.

The alternating-band experiment produced a camera-cluster mean absolute error of 0.186 in (95% camera-bootstrap interval, 0.156–0.218) in 2024 development cameras and 0.208 in (0.158–0.262) in locked 2025 cameras (Table 2; Figure 3). The locked interval overlaps the development interval, which argues against a large year-specific failure but does not establish equivalence with only three test cameras.

Correctly directed root extrapolation showed little aggregate signed bias but increasing dispersion with distance below the fitted bands. At 1 ft below the retained range, 1110 predictions from 10 cameras had an unweighted camera-mean bias of 0.40 cm, mean absolute error of 1.91 cm (95% camera-bootstrap interval, 1.24–2.98), and root mean squared error of 3.95 cm. Across all 1–3-ft

distances, unweighted camera-mean absolute error rose to 2.93 cm (1.86–4.64) and root mean squared error to 6.78 cm. One development camera contributed only five usable images and had much larger 3-ft error, visible in Figure 3; this is a failure condition rather than evidence for broadly reliable extrapolation.

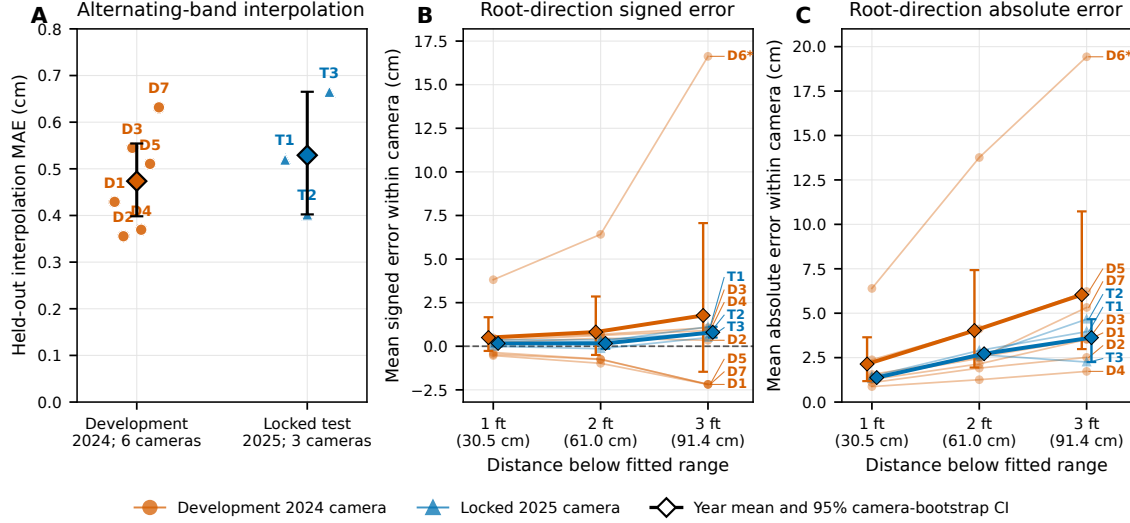


Figure 3: Camera-level physical-calibration validation. (a) Alternating-band interpolation mean absolute error for each 2024 development camera (D) and locked 2025 camera (T); diamonds and vertical intervals are unweighted year means and 95% whole-camera bootstrap intervals. (b) Mean signed error and (c) mean absolute error when lower bands are dropped, the retained upper bands are fit, and positions 1–3 ft below the fitted range are predicted. Thin lines are cameras; thick lines and intervals are unweighted camera means and whole-camera bootstrap intervals. D6, marked with an asterisk, has only five usable images. Individual bands are automatic candidates; the inferential unit is the camera.

Table 2: Held-out calibration performance. Interpolation intervals resample whole cameras. Root-direction rows give unweighted means of camera-level error summaries from the correctly directed upper-to-lower experiment.

Experiment	Cameras	Predictions	Bias	MAE	RMSE
				Physical error	
2024 alternating bands	6	1083	–	0.186 in	–
2025 alternating bands	3	427	–	0.208 in	–
1 ft below fitted range	10	1110	0.40 cm	1.91 cm	3.95 cm
1–3 ft below fitted range	10	2162	0.58 cm	2.93 cm	6.78 cm

3.2 Prior-guided candidate selection

In the controlled ambiguity experiment, the single-frame rule had plant-level height root mean squared error 10.48 cm (95% plant-bootstrap interval, 9.84–11.13), and post-detection filtering reduced it to 8.61 cm (8.07–9.16). Allowing the predictive height distribution to score candidates before the update reduced it further to 6.01 cm (5.47–6.57; Figure 4). The within-plant reduction relative to post-detection filtering was 2.60 cm (95% interval, 2.00–3.20). Prior-guided selection chose the truth-closest candidate 97.0% of the time. Benefits varied by simulated condition and were smallest when images were missing or growth plateaued early. This experiment establishes the candidate-choice mechanism under known ambiguity; it is not a claim that the same effect size occurred in the field data.

In the matched real-image audit, post-detection filtering and height-prior candidate scoring selected the same candidate in 100.0% of 63 evaluated plant-dates. Their mean absolute errors were therefore identical at 13.07 cm, compared with 13.38 cm for the single-frame candidate. The natural-image subset contained no candidate-choice conflict under the prespecified gates, so this audit is a negative control rather than evidence for a field accuracy gain from candidate re-ranking.

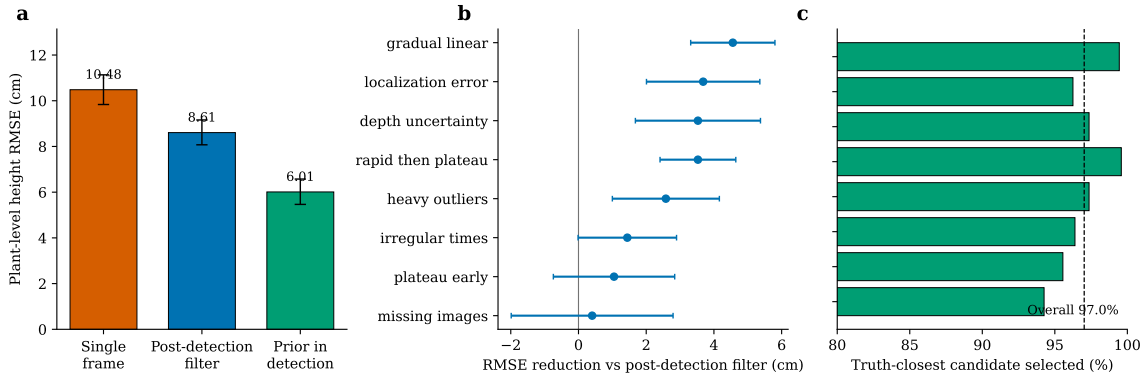


Figure 4: Controlled candidate-ambiguity experiment with 480 simulated plants across 8 scenarios. (a) Mean plant-level root mean squared error with 95% plant-bootstrap intervals. (b) Within-scenario reduction relative to post-detection filtering; points are means across 60 plants and bars are normal 95% intervals across plants. Positive values favor prior-guided candidate selection. (c) Percentage of frames in which prior-guided scoring selected the candidate closest to the known latent height. Every method saw the same candidates and used the same state dynamics; only the location at which the prior acted differed.

3.3 Independent physical-height validation

The primary independent validation used all 132 manual measurements obtained in 2021 from 33 plants across 12 stationary-camera rows and 5 dates. The test plants did not occur in the 2024–2025 development data, and manual heights were used only for evaluation; they did not enter the image measurement, growth field, filter, noise parameters, or comparator tuning. The single-frame image extent had signed error +2.64 cm, mean absolute error 12.03 cm, and root mean squared error 16.81 cm. The robust Bayesian posterior mean had corresponding values +2.73, 10.97, and 15.81 cm (Table 3; Figure 5). It had the lowest point-estimate error among the five fixed causal comparison methods. The within-record mean-absolute-error reduction relative to the single frame was 1.06 cm (95% row-cluster bootstrap interval, -0.16 to 2.39), and relative to the Gaussian local-linear Kalman filter it was 0.66 cm (-0.43 to 1.79). Both intervals include zero, so the observed advantage over the strongest comparators is modest and uncertain at this pilot scale. Against the three simpler longitudinal baselines, the corresponding gains were 4.01 cm (1.48–6.73) over EWMA, 7.80 cm (3.83–12.68) over the running median, and 3.08 cm (0.77–5.48) over Holt; all three row-cluster intervals excluded zero. The first image in a plant track cannot use preceding longitudinal information. In the mechanism-aligned secondary analysis of the 99 later records, mean absolute error decreased from 12.58 to 11.14 cm (gain, 1.44 cm; row-cluster interval, -0.21 to 3.26), retaining the direction of the all-record result with greater uncertainty. Across all records, median absolute error decreased from 8.74 to 7.30 cm; the 1.43-cm reduction had a whole-row bootstrap interval of 0.29 to 3.70 cm. The mean-error gain remained positive after omitting each camera row in turn, ranging from 0.61 to 1.42 cm across 12 omissions. These secondary distributional and influence summaries show that the direction is not attributable to one camera row or a few tail errors; they do not replace the primary mean-effect interval.

3.4 Physical-height uncertainty and temporal transfer

Against the independent 2021 manual heights, central 80% latent-height posterior intervals contained 115 of 132 records (87.1%; 95% whole-row bootstrap interval, 79.3–93.6%) and averaged 47.9 cm wide (46.9–49.4 cm). Central 95% intervals contained 125 of 132 records (94.7%; 89.4–98.6%) and averaged 75.4 cm wide (73.5–78.3 cm). Manual height was excluded from interval construction. These are posterior agreement intervals after the current image, using manual height as the physical reference

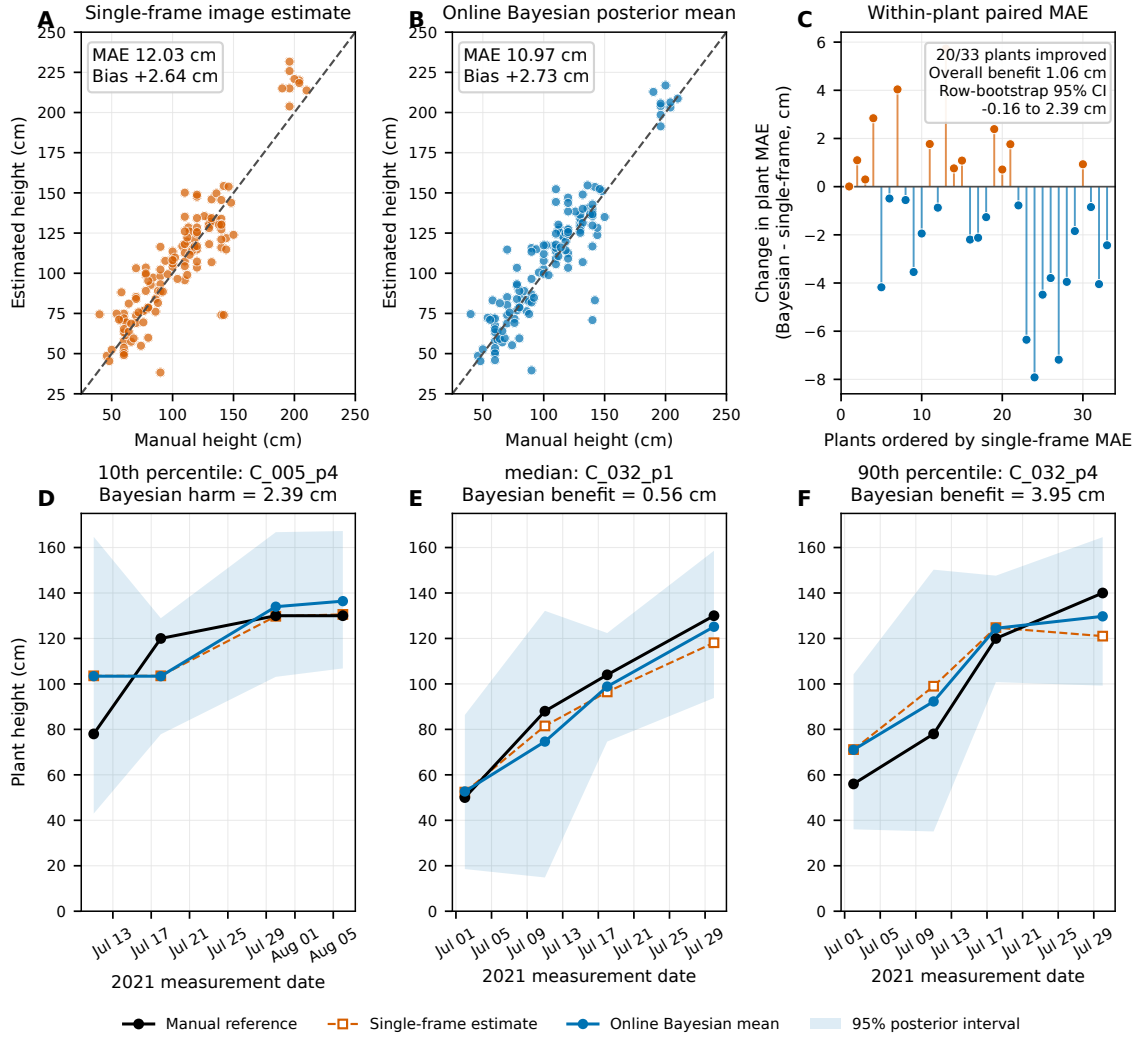


Figure 5: Independent 2021 manual-reference validation. (a,b) Single-frame and Bayesian estimates against manual height for all 132 records; dashed lines mark equality. (c) Within-plant difference in mean absolute error, ordered by single-frame error; negative values favor the Bayesian estimate. (d-f) Trajectories selected by the fixed 10th-, 50th-, and 90th-percentile rule on within-plant benefit, illustrating harm, median behavior, and benefit without residual-based exclusion. Shading is the 95% latent-height posterior interval. Within-record uncertainty is the stationary-camera-row bootstrap interval in panel c and Table 3.

Table 3: Independent physical-height validation in 2021. Every method is causal in image order and uses settings not tuned on manual height. The analysis includes all 132 plant-date records from 33 plants across 12 stationary-camera rows. Within-record method differences are evaluated by resampling whole camera rows.

Estimator	n	Signed error (cm)	MAE (cm)	RMSE (cm)
Single-frame image extent	132	+2.64	12.03	16.81
EWMA ($\alpha = 0.5$)	132	-9.78	14.98	20.16
Running median (3 images)	132	-12.15	18.77	27.14
Holt level-trend	132	-8.09	14.04	19.25
Gaussian local-linear Kalman	132	+2.74	11.62	16.52
Robust Bayesian particle filter	132	+2.73	10.97	15.81

proxy; they do not estimate error in the manual measurement itself.

As a secondary temporal transfer check, exact mixture-based predictive evaluation materially changed the interpretation of uncertainty (Table 4). Guessed noise settings covered 88.6% of next image-derived measurements with a mean 95% width of 66.6 cm. After the depth correction was applied to every centimeter-scale noise quantity, the 2024-derived model covered 96.4% with a 98.1-cm mean width over the full stream. On the temporally later, subject-disjoint 2025 subset, coverage was 94.9% with a 97.8-cm mean width across 372 forecasts from 11 plant tracks and 2 cameras. Its weighted interval score was 8.60 cm. Plants were newly grown in 2025, so no biological subject recurred from the 2024 development cohort; repeated forecasts within each plant and camera remain dependent observations. Lower nominal levels remained conservative (Supplementary Figure S3), so the near-nominal 95% result should not be interpreted as complete distributional calibration. The calibrated Bayesian predictive mean had mean absolute error 15.13 cm and root mean squared error 21.34 cm, compared with 16.64/23.63 cm for the last observation carried forward and 19.95/27.52 cm for two-point linear extrapolation. This two-camera check assesses temporal interval transfer to the next noisy image measurement; it is not the primary physical-height validation or evidence of broad camera generalization.

3.5 Prefix-causality audit

All 391 camera-time prefix recomputations matched the corresponding rows of the complete run, with 0 mismatches among 51,035 cumulative row comparisons. This verifies prefix invariance for the cached detection-to-measurement pipeline under fixed settings. It does not show that model development occurred prospectively, and it does not cover raw-image detector training. Sequential

Table 4: Uncertainty evaluations with distinct targets. Predictive intervals are formed before the next image-derived measurement; posterior intervals are formed after the current 2021 image and compared with manual field height.

Target	Configuration	n	Clusters	Coverage (%)	Mean width (cm)
Next image-derived measurement	Guessed noise, all years	1638	8	88.6	66.6
Next image-derived measurement	2024-derived, all years	1638	8	96.4	98.1
Next image-derived measurement	Secondary 2025 temporal test	372	2	94.9	97.8
2021 manual plant height	80% posterior agreement interval	132	12 rows	87.1	47.9
2021 manual plant height	95% posterior agreement interval	132	12 rows	94.7	75.4

position updating associated 2018 detections, compared with 1970 under frozen first-frame anchors, and reduced mean normalized association distance from 0.0396 to 0.0283. Because 2024–2025 manual plant heights are unavailable, these association results should be read as operational diagnostics rather than physical-height accuracy.

3.6 Human-annotated detector agreement

Five-fold detector validation matched 149 of 150 human masks (99.3%). The top keypoint was more difficult than the root (mean errors 28.3 and 18.2 pixels). Predicted vertical extent had an error of 29.5 pixels, corresponding under the recorded 2021 geometric scale to mean absolute error 12.27 cm, signed error +3.90 cm, and root mean squared error 15.91 cm; correlation with human mask extent was 0.837. These values replace the earlier centimeter conversion that had been fit using field height. They measure agreement with a human mask extent, not an independently clicked agronomic collar.

4 Discussion

This study develops a physically interpretable route from an accumulating longitudinal fixed-camera image stream to an as-of plant-height distribution after each image. The key methodological contribution is feedback across the usual image/model boundary: the predicted longitudinal state participates in current-image analysis before a measurement is committed. A running position state informs association and base refinement, the stronger candidate branch lets the predicted height distribution score alternative top detections, physical references convert the selected extent

382 to centimeters, and a robust particle filter updates plant-specific height, growth, and uncertainty.
383 Prior field-maize systems have reused earlier locations for later segmentation and height extraction
384 (Gao et al., 2022; Li et al., 2023), and plant vision has used Kalman tracking (Li et al., 2022). Our
385 contribution is the explicit prior-guided image-analysis loop inside a physically calibrated Bayesian
386 longitudinal method, with separate evidence for each executed branch.

387 The evidence is strongest for component behavior and weakest for a large field accuracy gain.
388 Repeated 2021 support-pole traces had a 1.24% median within-series coefficient of variation, supporting
389 temporal stability of the projected reference. Held-out red-band interpolation transferred from six
390 development cameras to three locked cameras with sub-inch error. The root-direction experiment also
391 revealed why interpolation accuracy cannot be extrapolated casually: errors widened rapidly as the
392 predicted point moved below the fitted band range, and one camera was particularly unstable. The
393 candidate simulation showed that a height prior can prevent a confident but implausible candidate
394 from entering the filter. The real-image audit, however, produced identical candidate choices for
395 the two temporal methods. The height-prior candidate mechanism therefore remains demonstrated
396 under controlled ambiguity rather than confirmed as a field advantage.

397 The primary physical-height evidence is the independent 2021 validation across 12 stationary-
398 camera rows, 33 plants, and five dates. Test plants were absent from the 2024–2025 development
399 data. Manual height was excluded from fitting and tuning, every matched record was retained, and
400 the bootstrap resampled whole camera rows to preserve dependence. Mean absolute error decreased
401 by about 1 cm relative to a single frame and by 0.66 cm relative to a Gaussian local-linear Kalman
402 filter, but both camera-row-cluster intervals included no improvement. In contrast, intervals for
403 the improvements over EWMA, running median, and Holt excluded zero. The mechanism-aligned
404 after-first-image analysis retained the direction of improvement but remained imprecise. Secondary
405 robustness checks showed a positive median-error reduction and a positive mean-error gain under
406 every leave-one-row-out omission, reducing concern that a single camera row or a few extreme errors
407 generated the direction of effect. They do not overturn the uncertainty in the primary mean-effect
408 interval. The 95% posterior interval covered 94.7% of the held-out manual heights, although its 75.4-
409 cm mean width shows substantial uncertainty. The results do not support a claim that longitudinal
410 Bayesian filtering uniformly improves height. They show that the robust particle filter achieved the
411 best point estimate across the fixed causal baselines and supplied uncertainty with near-nominal 95%
412 agreement without manual-height tuning.

413 The secondary temporal uncertainty check answers a different question. Applying the depth
 414 conversion to the observation, process, and outlier scales corrected a consequential unit mismatch.
 415 With parameters estimated from 2024 only, 95% coverage was 94.9% on the temporally later,
 416 subject-disjoint 2025 cohort with a 97.8-cm mean width, while 50%, 80%, and 90% intervals
 417 overcovered. Because the test contains only two cameras, it supports year-forward interval transfer
 418 for those installations rather than broad cross-camera calibration. Repeated same-time images,
 419 direct uncertainty labels, or a hierarchical design are needed to separate measurement and biological
 420 variation and narrow intervals without losing coverage.

421 Several limitations bound the conclusions. First, the primary independent validation spans 12
 422 stationary-camera rows but only 33 plants in one 2021 season. The secondary 2025 interval test has
 423 11 plant tracks at two cameras; the locked calibration test has three cameras. Second, the 2024–2025
 424 stream lacks manual plant-height outcomes. The independent 2021 study supplies physical-height
 425 truth on five dates through season end, using archived segmentation extents and an external growth
 426 field estimated without 2021 labels. Third, human validation serves component-specific purposes:
 427 150 plant masks assess image extent, 132 field measurements assess physical height, and Haoming
 428 Wang’s 199 traces of 61 2021 images audit physical-reference geometry. These sources do not estimate
 429 inter-reader variation for agronomic landmarks. Fourth, the depth ratio assumes known distances,
 430 level ground, negligible roll, and stable camera geometry. Fifth, the released checkpoint permits
 431 inference on the curated images, but its complete original training provenance remains only partially
 432 documented. Sixth, source-frame matching recovered positions for 87 archived plant crops; 63 map to
 433 the primary validation across six camera rows. For the remaining 69 records, the archive preserves the
 434 derived segmentation extent but not a reconstructable raw-image-to-extent chain. These boundaries
 435 are addressed with label withholding, subject-disjoint years, camera-row inference, negative controls,
 436 and explicit provenance, but they limit generalization beyond the evaluated fields and camera layouts.

437 For field use, the posterior width is as consequential as the point estimate. The broad intervals
 438 here can flag images or installations requiring review, while the documented failure of downward
 439 extrapolation cautions against measurements outside the calibrated reference range. The method is
 440 most useful where the same fixed view is accumulated through time and where association or candidate
 441 ambiguity can be resolved by biologically plausible temporal information. The open candidate tables
 442 and per-record outputs allow future users to audit each decision rather than accepting only a final
 443 growth curve.

5 Conclusions

Prior-guided image analysis connects accumulating images to measurement extraction by allowing a Bayesian longitudinal state to inform current plant association and, when alternatives exist, candidate choice. With physical-reference calibration and robust sequential updating, the method provided the lowest MAE among six causal estimators on 132 held-out 2021 manual-height records. Row-cluster intervals supported improvements over EWMA, running median, and Holt; differences from the single-frame and Gaussian filters remained uncertain. The 95% posterior coverage was near nominal but wide, and year-forward 2025 intervals transferred in a two-camera test. The controlled ambiguity experiment confirmed the prior-in-image mechanism, whereas the matched field subset contained no natural candidate conflict. Released code, images, annotations, checkpoint, candidate cache, and derived data make these claims auditable. The evidence supports a transparent, accessible workflow for longitudinal maize height phenotyping and a testable architecture for using temporal state within image analysis.

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Author Contributions

Haoming Wang: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. Chong Wang: Conceptualization, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. Yawei Li: Data curation, Investigation, Validation, Writing – review and editing. Cheng-Ting Yeh: Validation, Writing – review and editing. Patrick S. Schnable: Conceptualization, Funding acquisition, Resources, Supervision, Writing – review and editing. Peng Liu: Conceptualization, Formal analysis, Methodology, Project administration, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review and editing. All authors approved the final manuscript.

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474 **Conflict of Interest**

475 The authors declare no conflict of interest.

476 **Data Availability Statement**

477 Analysis code, the 61 curated 2021 images, 12 human-annotated support-pole XML files, the imported
478 199-trace audit table, the released pose checkpoint, candidate predictions, derived data, figure sources,
479 and manuscript source are available in the versioned public release. The repository also provides
480 the 87-record partial source-frame coordinate recovery described above, an MIT license for code,
481 and a SHA-256 manifest. No separate reuse license has been confirmed for non-code materials. The
482 larger 2024–2025 raw-image archive remains in institutional storage; checksum-identified derived
483 measurements needed for the reported analyses are public.

484 **Declaration of Generative AI Use**

485 Generative AI tools (Claude, Anthropic, and Codex, OpenAI) assisted data exploration, code review,
486 figure and manuscript preparation, and language editing under direct author supervision. All
487 numerical results were produced by the included analysis code, and the authors take responsibility
488 for the design, verification, interpretation, and final text.

489 **Supplementary Materials**

490 Materials and Methods S1. Detailed model, causal-update, calibration, and reproducibility specifica-
491 tion. Figure S1. Pinhole geometry and reference-to-target depth correction. Figure S2. Example
492 posterior trajectories. Figure S3. Multi-level predictive-interval calibration. Table S1. Full parameter
493 and implementation specification. Table S2. Predictive-interval calibration. Table S3. Real-image
494 candidate audit. Table S4. Field-error distribution and row-influence sensitivity. Table S5. Per-plant
495 mature-date field comparison. Table S6. Data and result provenance. Data S1. Curated 2021 images

and human support-pole annotations. Data S2. Canonical derived outputs and audit files. A separate Supplementary Materials PDF accompanies this manuscript.

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